



Protocol Audit Report

Version 1.0

CyberGhost

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PasswordStore Audit Report

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Protocol Summary

The PasswordStore protocol is designed as a simple vault intended to allow its deployer (the contract owner) to securely store and update a single private text password on the Ethereum blockchain. Upon deployment, the contract assigns the deployer's address as the owner, who is intended to be the sole party capable of setting new password values and retrieving the stored password via an access-controlled getter function. Intended as a personal password manager on-chain, its core objective is to ensure that confidential credentials remain restricted exclusively to the contract owner while allowing them to update the password whenever needed.

Disclaimer

CyberGhost makes all effort to find as many vulnerabilities in the code in the given time period, but holds no responsibilities for the findings provided in this document. A security audit by the team is not an endorsement of the underlying business or product. The audit was time-boxed and the review of the code was solely on the security aspects of the Solidity implementation of the contracts.

Risk Classification

		Impact		
		High	Medium	Low
Likelihood	High	H	H/M	M
	Medium	H/M	M	M/L
	Low	M	M/L	L

We use the CodeHawks severity matrix to determine severity. See the documentation for more details.

Audit Details

The findings described in this document correspond the following commit hash:

```
1 7d55682ddc4301a7b13ae9413095feffd9924566
```

Scope

```
1 ./src/  
2 #-- PasswordStore.sol
```

Roles

- Owner: The user who can set the password and read the password.
- Outsiders: No one else should be able to set or read the password.

Executive Summary

CyberGhost audited the PasswordStore protocol, uncovering 3 issues (2 High, 1 Info).

The High-severity flaws completely compromise the protocol: any user can read the raw password off-chain via RPC storage reads, and unauthenticated addresses can overwrite the owner's password due to missing access controls. Immediate remediation is required before deployment.

Issues found

Severity	Number of issues found
High	2
Medium	0
Low	0
Info	1
Total	3

Recommended Mitigation: Sensitive data such as passwords should never be stored in plain text on-chain. If the protocol requires off-chain retrieval of the original password, the data must be encrypted off-chain using asymmetric encryption (e.g., ECIES) or symmetric encryption (e.g., AES) before submitting it to the contract. The smart contract should only store the encrypted ciphertext. Authorized users can then fetch this ciphertext via `PasswordStore::getPassword` and decrypt it off-chain using their private key.

[H-2] PasswordStore::setPassword has no access control, meaning a non-owner could change the password.

Description: The `PasswordStore::setPassword` function is set to be an `external` function, however, the `natSpec` of the function and overall purpose of the smart contract is that: `This function allows only the owner to set a new password.`

```
1 function setPassword(string memory newPassword) external {
2   @> // @audit EVERYONE is able to change the owner's password.
3     s_password = newPassword;
4     emit SetNetPassword();
5 }
```

Impact: Anyone can set/change the contract's password, severely breaking the contract intended functionality.

Proof of Concept: Add the following to the `PasswordStore.t.sol` test file:

Code

```
1 function test_anyone_can_set_password(address randomAddress) public {
2   vm.assume(randomAddress != owner);
3   vm.prank(randomAddress);
4   string memory expectedPassword = "myNewPassword";
5   passwordStore.setPassword(expectedPassword);
6
7   vm.prank(owner);
8   string memory actualPassword = passwordStore.getPassword();
9   assertEq(actualPassword, expectedPassword);
10 }
```

Recommended Mitigation: Add an access control to the `PasswordStore::setPassword` function:

```
1 if(msg.sender != owner) {
2   revert PasswordStore__NotOwner();
3 }
```

Informational

[I-1] The PasswordStore::getPassword natSpec indicates a parameter that doesn't exist, causing the natSpec to be incorrect.

Description:

```
1  /*
2  * @param newPassword The new password to set.
3  * @audit There is no newPassword parameter.
4  */
5  function getPassword() external view returns (string memory);
```

The `PasswordStore::getPassword` function signature is `getPassword()` while the natSpec says it should be `getPassword(string)`.

Impact: The natSpec is incorrect.

Recommended Mitigation: Remove the incorrect natSpec line.

```
1  - * @param newPassword The new password to set.
```